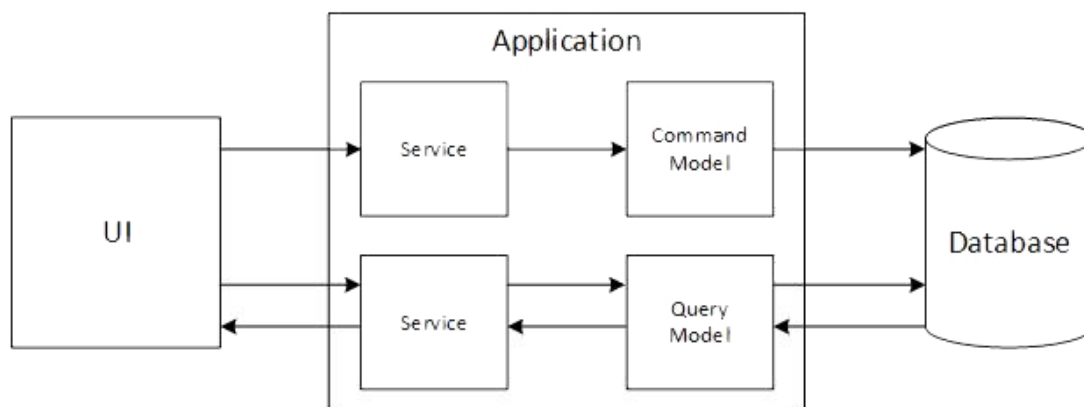


The query model and the command model

One way to overcome the challenges of using a single object model for both queries and commands is to separate the two. This pattern, known as CQRS, results in two separate models. The **query model** is responsible for reads and the **command model** is responsible for updates:



Query objects only return data and do not alter state, while command objects alter state and do not return data. Another way of looking at this concept is that asking a question (a query) should not alter the answer. In order to perform an action that will mutate state, a command is used.

When the system needs to read data, it goes through the query model, and when the system needs to update data, it goes through the command model. As part of processing a command, the system may need to read data, but beyond what is necessary for completing a command, the reading of data should go through the query model.

Although not required, if CQRS is taken to the next level, the query and command models can be made to utilize separate databases. This gives each of the two models its own schema, which can be optimized for its specific usage. If separate databases are used, the two databases must be kept in sync. One way to do that is through the use of events.